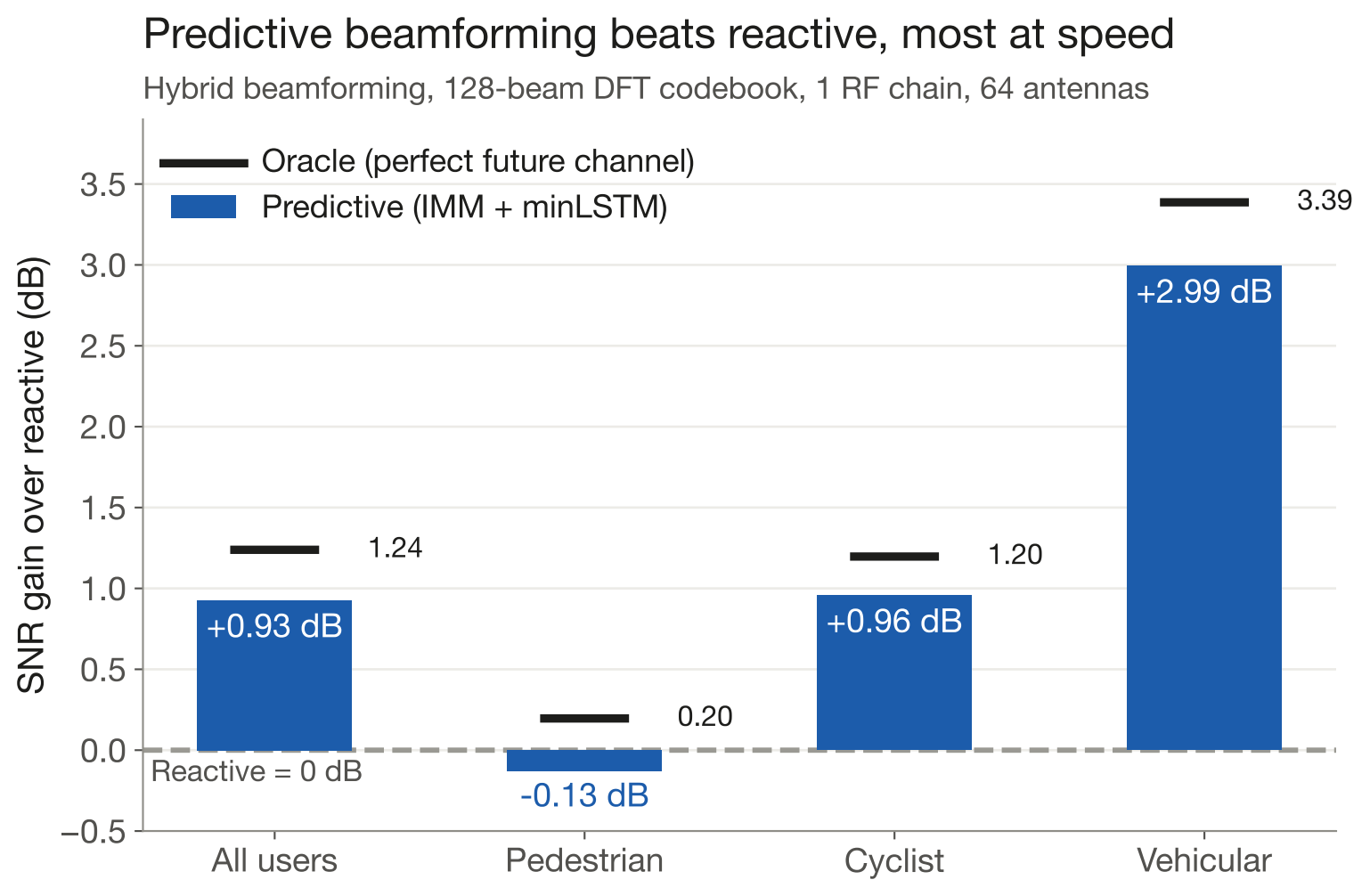
Predictive beamforming beats reactive, most at speed
Hybrid beamforming, 128-beam DFT codebook, 1 RF chain, 64 antennas
Oracle (perfect future channel)
Predictive (IMM + minLSTM)
SNR gain over reactive (dB)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
1.24
+0.93 dB
0.20
-0.13 dB
1.20
+0.96 dB
3.39
+2.99 dB
Reactive = 0 dB
All users
Pedestrian
Cyclist
Vehicular